

CS6310 – Software Architecture & Design

Assignment #3 [100 points]: Pokémon Tournament Project – Phase 3 (v0)

"Welcome to the Poke-Dome!"

Spring 2025 Semester – created by the CS6310 Instructional Team

Tournament Test Case Walkthrough

This is an example of the tournament test case results, with expected output lines interspersed. This can be very helpful when clarifying the details of how the system should behave based on the updated core functionality.

Welcome to the thunder dome!

The **"Welcome to the thunder dome!"** message must be displayed at the beginning of each simulation run

```
tournament,Pikachu,Bulbasaur,Charmander,Squirtle  
> tournament,Pikachu,Bulbasaur,Charmander,Squirtle
```

Note that the ordering means that Pikachu and Bulbasaur must battle in the 1st round, followed by Charmander and Squirtle in the 2nd round. Then, the winner of each of the first two rounds will battle in the 3rd round to determine the winner. A tournament is an ordering of a sequence of battles.

Starting tournament round 1 with Pikachu and Bulbasaur

```
Pikachu is attacking with Thunder Shock for 3 damage to Bulbasaur  
Bulbasaur has received 3 dmg, remaining hp is 22  
Bulbasaur is attacking with Leaf Storm for 6 damage to Pikachu  
Pikachu has received 6 dmg, remaining hp is 19  
Pikachu is attacking with Thunder for 6 damage to Bulbasaur  
Bulbasaur has received 6 dmg, remaining hp is 16  
Bulbasaur is attacking with Tackle for 1 damage to Pikachu  
Pikachu has received 1 dmg, remaining hp is 18  
Pikachu is attacking with Tail Whip for 2 damage to Bulbasaur  
Bulbasaur has received 2 dmg, remaining hp is 14  
Bulbasaur is attacking with Tackle for 1 damage to Pikachu  
Pikachu has received 1 dmg, remaining hp is 17  
Pikachu is attacking with Thunder for 6 damage to Bulbasaur  
Bulbasaur has received 6 dmg, remaining hp is 8  
Bulbasaur is attacking with Vine Whip for 2 damage to Pikachu  
Pikachu has received 2 dmg, remaining hp is 15  
Pikachu is attempting to defend with Endure  
Bulbasaur is attempting to defend with Endure  
Pikachu is attacking with Thunder Shock for 3 damage to Bulbasaur  
Bulbasaur successfully reduced Pikachu's damage by 1 with Endure  
Bulbasaur has received 2 dmg, remaining hp is 6  
Bulbasaur is attempting to defend with Protect  
Pikachu is attacking with Tail Whip for 2 damage to Bulbasaur  
Bulbasaur successfully reduced Pikachu's damage by 3 with Protect  
Bulbasaur has received 0 dmg, remaining hp is 6  
Bulbasaur is attempting to defend with Protect  
Pikachu is attempting to defend with Block
```

Bulbasaur is attacking with Vine Whip for 2 damage to Pikachu
Pikachu successfully reduced Bulbasaur's damage by 2 with Block
Pikachu has received 0 dmg, remaining hp is 15
Pikachu is attempting to defend with Endure
Bulbasaur is attempting to defend with Block
Pikachu is attempting to defend with Block
Bulbasaur is attempting to defend with Block
Pikachu is attacking with Thunder for 6 damage to Bulbasaur
Bulbasaur successfully reduced Pikachu's damage by 2 with Block
Bulbasaur has received 4 dmg, remaining hp is 2
Bulbasaur is attacking with Tackle for 1 damage to Pikachu
Pikachu has received 1 dmg, remaining hp is 14
Pikachu is attacking with Growl for 1 damage to Bulbasaur
Bulbasaur has received 1 dmg, remaining hp is 1
Bulbasaur is attempting to defend with Block
Pikachu is attempting to defend with Block
Bulbasaur is attempting to defend with Endure
Pikachu is attacking with Growl for 1 damage to Bulbasaur
Bulbasaur successfully reduced Pikachu's damage by 1 with Endure
Bulbasaur has received 0 dmg, remaining hp is 1
Bulbasaur is attempting to defend with Block
Pikachu is attempting to defend with Protect
Bulbasaur is attacking with Leaf Storm for 6 damage to Pikachu
Pikachu successfully reduced Bulbasaur's damage by 3 with Protect
Pikachu has received 3 dmg, remaining hp is 11
Pikachu is attempting to defend with Endure
Bulbasaur is attacking with Razor Leaf for 3 damage to Pikachu
Pikachu successfully reduced Bulbasaur's damage by 1 with Endure
Pikachu has received 2 dmg, remaining hp is 9
Pikachu is attempting to defend with Block
Bulbasaur is attempting to defend with Protect
Pikachu is attacking with Thunder for 6 damage to Bulbasaur
Bulbasaur successfully reduced Pikachu's damage by 3 with Protect
Bulbasaur has received 3 dmg, remaining hp is 0
Bulbasaur has lost
Pikachu has won the battle
Pikachu has won round 1

The 1st round battle has been concluded, and now the 2nd round battle commences...

Starting tournament round 2 with Charmander and Squirtle
Charmander is attacking with Flamethrower for 6 damage to Squirtle
Squirtle has received 6 dmg, remaining hp is 19
Squirtle is attacking with Hydro Pump for 6 damage to Charmander
Charmander has received 6 dmg, remaining hp is 19
Charmander is attacking with Flamethrower for 6 damage to Squirtle
Squirtle has received 6 dmg, remaining hp is 13
Squirtle is attempting to defend with Protect
Charmander is attacking with Scratch for 2 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 3 with Protect

Squirtle has received 0 dmg, remaining hp is 13
Squirtle is attempting to defend with Block
Charmander is attacking with Ember for 3 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 2 with Block
Squirtle has received 1 dmg, remaining hp is 12
Squirtle is attacking with Water Gun for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 16
Charmander is attacking with Ember for 3 damage to Squirtle
Squirtle has received 3 dmg, remaining hp is 9
Squirtle is attempting to defend with Block
Charmander is attempting to defend with Protect
Squirtle is attacking with Attack for 1 damage to Charmander
Charmander successfully reduced Squirtle's damage by 3 with Protect
Charmander has received 0 dmg, remaining hp is 16
Charmander is attempting to defend with Endure
Squirtle is attempting to defend with Endure
Charmander is attempting to defend with Endure
Squirtle is attacking with Tackle for 2 damage to Charmander
Charmander successfully reduced Squirtle's damage by 1 with Endure
Charmander has received 1 dmg, remaining hp is 15
Charmander is attempting to defend with Protect
Squirtle is attacking with Attack for 1 damage to Charmander
Charmander successfully reduced Squirtle's damage by 3 with Protect
Charmander has received 0 dmg, remaining hp is 15
Charmander is attempting to defend with Block
Squirtle is attacking with Attack for 1 damage to Charmander
Charmander successfully reduced Squirtle's damage by 2 with Block
Charmander has received 0 dmg, remaining hp is 15
Charmander is attacking with Scratch for 2 damage to Squirtle
Squirtle has received 2 dmg, remaining hp is 7
Squirtle is attempting to defend with Protect
Charmander is attempting to defend with Endure
Squirtle is attempting to defend with Protect
Charmander is attacking with Ember for 3 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 3 with Protect
Squirtle has received 0 dmg, remaining hp is 7
Squirtle is attempting to defend with Block
Charmander is attacking with Ember for 3 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 2 with Block
Squirtle has received 1 dmg, remaining hp is 6
Squirtle is attempting to defend with Block
Charmander is attempting to defend with Block
Squirtle is attempting to defend with Block
Charmander is attacking with Attack for 1 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 2 with Block
Squirtle has received 0 dmg, remaining hp is 6
Squirtle is attempting to defend with Block
Charmander is attempting to defend with Protect
Squirtle is attempting to defend with Endure
Charmander is attacking with Ember for 3 damage to Squirtle

Squirtle successfully reduced Charmander's damage by 1 with Endure
Squirtle has received 2 dmg, remaining hp is 4
Squirtle is attacking with Water Gun for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 12
Charmander is attempting to defend with Protect
Squirtle is attempting to defend with Endure
Charmander is attacking with Ember for 3 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 1 with Endure
Squirtle has received 2 dmg, remaining hp is 2
Squirtle is attacking with Water Gun for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 9
Charmander is attempting to defend with Block
Squirtle is attempting to defend with Protect
Charmander is attempting to defend with Protect
Squirtle is attacking with Water Gun for 3 damage to Charmander
Charmander successfully reduced Squirtle's damage by 3 with Protect
Charmander has received 0 dmg, remaining hp is 9
Charmander is attempting to defend with Endure
Squirtle is attempting to defend with Endure
Charmander is attacking with Flamethrower for 6 damage to Squirtle
Squirtle successfully reduced Charmander's damage by 1 with Endure
Squirtle has received 5 dmg, remaining hp is 0
Squirtle has lost
Charmander has won the battle
Charmander has won round 2

And finally, Pikachu and Charmander face each other in the 3rd and final round. Note that they must both be restored to their full health (i.e., 20 points) before commencing the final battle.

Starting tournament round 3 with Pikachu and Charmander
Pikachu is attacking with Thunder for 6 damage to Charmander
Charmander has received 6 dmg, remaining hp is 19
Charmander is attacking with Attack for 1 damage to Pikachu
Pikachu has received 1 dmg, remaining hp is 24
Pikachu is attacking with Thunder Shock for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 16
Charmander is attempting to defend with Block
Pikachu is attacking with Thunder for 6 damage to Charmander
Charmander successfully reduced Pikachu's damage by 2 with Block
Charmander has received 4 dmg, remaining hp is 12
Charmander is attacking with Flamethrower for 6 damage to Pikachu
Pikachu has received 6 dmg, remaining hp is 18
Pikachu is attacking with Thunder Shock for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 9
Charmander is attempting to defend with Protect
Pikachu is attacking with Tail Whip for 2 damage to Charmander
Charmander successfully reduced Pikachu's damage by 3 with Protect
Charmander has received 0 dmg, remaining hp is 9
Charmander is attacking with Scratch for 2 damage to Pikachu
Pikachu has received 2 dmg, remaining hp is 16
Pikachu is attacking with Tail Whip for 2 damage to Charmander

Charmander has received 2 dmg, remaining hp is 7
Charmander is attempting to defend with Protect
Pikachu is attempting to defend with Block
Charmander is attempting to defend with Endure
Pikachu is attempting to defend with Endure
Charmander is attempting to defend with Protect
Pikachu is attempting to defend with Endure
Charmander is attempting to defend with Endure
Pikachu is attempting to defend with Endure
Charmander is attempting to defend with Block
Pikachu is attacking with Thunder Shock for 3 damage to Charmander
Charmander successfully reduced Pikachu's damage by 2 with Block
Charmander has received 1 dmg, remaining hp is 6
Charmander is attempting to defend with Endure
Pikachu is attempting to defend with Block
Charmander is attacking with Flamethrower for 6 damage to Pikachu
Pikachu successfully reduced Charmander's damage by 2 with Block
Pikachu has received 4 dmg, remaining hp is 12
Pikachu is attempting to defend with Block
Charmander is attempting to defend with Protect
Pikachu is attempting to defend with Endure
Charmander is attacking with Scratch for 2 damage to Pikachu
Pikachu successfully reduced Charmander's damage by 1 with Endure
Pikachu has received 1 dmg, remaining hp is 11
Pikachu is attempting to defend with Protect
Charmander is attempting to defend with Block
Pikachu is attacking with Thunder for 6 damage to Charmander
Charmander successfully reduced Pikachu's damage by 2 with Block
Charmander has received 4 dmg, remaining hp is 2
Charmander is attacking with Attack for 1 damage to Pikachu
Pikachu has received 1 dmg, remaining hp is 10
Pikachu is attacking with Growl for 1 damage to Charmander
Charmander has received 1 dmg, remaining hp is 1
Charmander is attempting to defend with Protect
Pikachu is attempting to defend with Endure
Charmander is attempting to defend with Block
Pikachu is attempting to defend with Endure
Charmander is attempting to defend with Block
Pikachu is attacking with Tail Whip for 2 damage to Charmander
Charmander successfully reduced Pikachu's damage by 2 with Block
Charmander has received 0 dmg, remaining hp is 1
Charmander is attacking with Ember for 3 damage to Pikachu
Pikachu has received 3 dmg, remaining hp is 7
Pikachu is attacking with Thunder Shock for 3 damage to Charmander
Charmander has received 3 dmg, remaining hp is 0
Charmander has lost
Pikachu has won the battle
Pikachu has won round 3
Pikachu has won the tournament

Pikachu wins! Pikachu wins! Note that your system must display the proper messages at the start & end of the rounds and the tournament.

> // battle between Bulbasaur & Pikachu

> battle,Bulbasaur,Pikachu

This command initiates a battle between Bulbasaur as <Pokemon_1> and Pikachu as <Pokemon_2>. From a functionality perspective, Bulbasaur must be assigned 10 for its randomized seed, and Pikachu must be assigned a randomized seed of 11. Bulbasaur will have the first attack opportunity.

Bulbasaur is attacking with Tackle for 1 damage to Pikachu

Pikachu has received 1 dmg, remaining hp is 24

Once the battle has started, the Pokémon must take turns and communicate with each other as they attack and defend. Bulbasaur gets the first turn to attack or defend itself. Bulbasaur decides to attack and uses the Tackle attack – causing 1 hit point of damage – against Pikachu. Since this is the first action of the battle, Pikachu hasn't had an opportunity to defend itself, so the full damage value from the attack is inflicted on Pikachu. Both Pokémon begin with twenty-five (25) hit points, so Pikachu's current hit points are reduced to 24. Now, it's Pikachu's turn.

Pikachu is attacking with Growl for 1 damage to Bulbasaur

Bulbasaur has received 1 dmg, remaining hp is 24

Now Pikachu gets the opportunity to attack or defend, and decides to use the Growl attack, causing 1 hit point of damage against Bulbasaur. Since Bulbasaur chose to attack on its previous turn, and not defend itself, it will receive the full damage value from Pikachu's attack. Bulbasaur's hit points are reduced from 25 to 24. Now, it's Bulbasaur's turn.

Bulbasaur is attempting to defend with Endure

Pikachu is attempting to defend with Endure

Bulbasaur is attempting to defend with Block

Pikachu is attempting to defend with Endure

These actions represent four consecutive turns where Bulbasaur and Pikachu both decided to defend themselves. No damage occurs to either of them without an attack. Now, it's Bulbasaur's turn.

Bulbasaur is attacking with Vine Whip for 2 damage to Pikachu

Pikachu successfully reduced Bulbasaur's damage by 1 with Endure

Pikachu has received 1 dmg, remaining hp is 23

Bulbasaur has decided to use the Vine Whip attack, which normally causes 2 hit points of damage. Pikachu, however, used the Endure defense on its previous turn, so this reduces the damage by one point. Consequently, Pikachu's hit points are reduced by $(2 - 1) = 1$ point, from 24 down to 23. Now, it's Pikachu's turn.

Pikachu is attempting to defend with Protect

Bulbasaur is attempting to defend with Block

Pikachu is attempting to defend with Protect

Bulbasaur is attempting to defend with Protect

Pikachu is attempting to defend with Endure

Bulbasaur is attempting to defend with Endure

More defensive posturing from both Pokémon. Now, it's Pikachu's turn.

Pikachu is attacking with Thunder Shock for 3 damage to Bulbasaur
Bulbasaur successfully reduced Pikachu's damage by 1 with Endure
Bulbasaur has received 2 dmg, remaining hp is 22

Pikachu has decided to use the Thunder Shock attack, which normally causes 3 hit points of damage. Bulbasaur, however, used the Endure defense on its previous turn, so this reduces the damage by one point. Consequently, Pikachu's hit points are reduced by $(3 - 1) = 2$ points, from 24 down to 22.

...

Pikachu is attempting to defend with Protect

We resume the battle many turns later, with Bulbasaur at 2 hit points, and Pikachu at 10 hit points. It was Pikachu's turn, and Pikachu defended itself with Protect. Now, it's Bulbasaur's turn.

Bulbasaur is attacking with Razor Leaf for 3 damage to Pikachu
Pikachu successfully reduced Bulbasaur's damage by 3 with Protect
Pikachu has received 0 dmg, remaining hp is 10

Bulbasaur has decided to use the Razor Leaf attack, which normally causes 3 hit points of damage. Pikachu, however, used the Protect defense on its previous turn, so this reduces the damage by three points. Consequently, Pikachu's hit points aren't reduced at all. Now, it's Pikachu's turn.

Pikachu is attacking with Tail Whip for 2 damage to Bulbasaur
Bulbasaur has received 2 dmg, remaining hp is 0
Bulbasaur has lost
Pikachu has won the battle

Pikachu attacks Bulbasaur with a Tail Whip causing 2 hit points of damage. Bulbasaur didn't defend itself last turn, so the full 2 hit points from the attack impacts Bulbasaur, and its hit points are reduced to zero. Therefore, Pikachu wins the battle.

```
// remove the seed  
> // remove the seed
```

The lines that begin with the ">" character are either comments or commands from the test case. Comments are indicated with the "///" delimiter and included in the test case to add clarity. This shows the input line starting with "///" and the output line starting with ">"

```
setseed,1  
> setseed,1
```

The **setseed,1** command sets the random seed value to 1 initially. This will be preserved until removed or the application stops.

```
setseed,10  
> setseed,10
```

This (second) **setseed,10** command sets the random seed value to 10. The line above it would be an out

removeseed

> removeseed

The **removeseed** command removes the effect of setting the seed to the value 1. This would allow random to be random.

stop

> stop

stop acknowledged

simulation terminated

The stop command ends the simulation loop gracefully.

Closing Comments & Suggestions

This is the information provided by the customer so far. We (the OMSCS 6310 Team) will conduct Office Hours where you will be permitted to ask us questions to clarify the client's intent, etc. We will answer some of the questions, but we will not necessarily answer all of them. Also, though this current version of the system is the "core" of the system going forward, our clients will add, update, and remove some of the requirements over the span of the course. One of your main tasks will be to ensure that your architectural documents and related artifacts remain consistent with the problem requirements – and with your system implementations – over time.

Quick Reminder on Collaborating with Others

Please use Ed Discussion for your questions and/or comments, and post publicly whenever it is appropriate. If your questions or comments contain information that specifically provides an answer for some part of the assignment, then please make your post private first, and we (the OMSCS 6310 Team) will review it and decide if it is suitable to be shared with the larger class. Best of luck to you with this assignment, and please contact us if you have questions or concerns.